

laser beams, and the longitudinal interval size is defined as a size of an interval between adjacent two of the recesses in the moving direction.

[0013] According to the present invention, the following configuration is performed: in the former phase of irradiating the laser beam on the surface of the member, irradiation intensity is quickly increased, and then, the increase rate of irradiation intensity per unit time is gradually reduced, whereas in the latter phase of irradiation, irradiation intensity is quickly reduced. Accordingly, as shown in FIG. 3, the recess is formed on the surface of the member in the irradiation position of the laser beam such that the middle part thereof is deeply recessed as a bottom. Moreover, the recess is shaped to have a cross section gently slanting toward the bottom.

[0014] Thus, with the configuration that the recess is formed on the surface of the member while being shaped to have a cross section including a slope gently slanting toward the bottom, even in applying adhesive with a certain high level of viscosity onto the surface of the member, the recess enables the adhesive to flow to the bottom more easily than, for instance, a recess formed in use of a square-wave laser beam (see FIG. 4) so as to be shaped to have a cross section having an approximately rectangular shape (see FIG. 5). Because of this, adhesiveness on the surface of the member can be enhanced.

[0015] Besides, the increase rate of irradiation intensity per unit time is gradually reduced in the former phase of irradiation, whereby it is possible to avoid sudden rise of current, i.e., occurrence of current spike. Hence, even in use of a laser diode as a light source of the laser beam, the laser diode can be mitigated in damage.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Referring now to the attached drawings which form a part of this original disclosure:

FIG. 1 is a diagram showing a laser irradiation device 10 according to an exemplary embodiment to which the present invention is applied;

FIG. 2 is a chart showing a waveform of a laser beam L in the exemplary embodiment to which the present invention is applied;

FIG. 3 is a cross-sectional view of a member S showing the shape of a recess X formed by the laser beam L in the exemplary embodiment to which the present invention is applied;

FIG. 4 is a chart showing a waveform of a square-wave laser beam;

FIG. 5 is a cross-sectional view of the member S showing a recess formed by the square-wave laser beam;

FIG. 6 is a diagram of the laser irradiation device 10 according to modification 1;

FIG. 7 is a plan view of the member S showing a large number of recesses X formed by the laser irradiation device 10 according to modification 1; and

FIG. 8 is a chart showing a relation between longitudinal interval size and adhesion strength.

DETAILED DESCRIPTION OF EMBODIMENTS

(Configuration of Laser Irradiation Device 10)

[0017] A configuration of a laser irradiation device 10 to which the present invention is applied will be hereinafter explained with reference to drawings. It should be noted that the laser irradiation device 10, to which the present invention is applied, may be used in, for instance, treatment for roughening a surface T of a CFRP (Carbon Fiber Reinforced Plastic) member S as with the present exemplary embodiment, and may be used in treatment for roughening the surface of a member made of not only CFRP but also metal or so forth.

[0018] As shown in FIG. 1, the laser irradiation device 10 mainly includes a laser light source 12 and a light collector 14.

[0019] The laser light source 12 is a member for generating a laser beam L with a predetermined wavelength and a predetermined waveform. In the present exemplary embodiment, a laser diode (semiconductor laser) is used as the laser light source 12. Obviously, the laser light source 12 is not limited to this; for instance, a laser processor capable of generating the laser beam L with higher power may be used.

[0020] The light collector 14 is a member for concentrating the laser beam L irradiated from the laser light source 12 on a predetermined focal position F. In the present exemplary embodiment, the light collector 14 is formed by combination